Answers From Chatgpt

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1 Explanation of the SU(2) Global Anomaly Remark

We explain the logic of the remark step by step.

1.1 SU(2) Gauge Theory in Four Dimensions

The weak interaction is described by an $SU(2)$ gauge theory. Therefore, on our four-dimensional spacetime M there is an $SU(2)$ principal bundle

$$P \to M.$$

On a local patch we may treat spacetime as $\mathbb{R}^4$, and a gauge transformation is a map

$$g : \mathbb{R}^4 \to SU(2).$$

1.2 Compactification to S^4 and Large Gauge Transformations

Assume the gauge transformation becomes trivial at spatial infinity:

$$g(x) \to \mathbf{1} \quad \text{as } |x| \to \infty.$$

This allows us to compactify $\mathbb{R}^4$ by adding a point at infinity:

$$\mathbb{R}^4 \cup \{\infty\} \cong S^4,$$

with $g(\infty) = \mathbf{1}$. Hence gauge transformations are classified by maps

$$g : S^4 \to SU(2) \simeq S^3.$$

The homotopy classes of such maps are elements of

$$\pi_4(S^3) \simeq \mathbb{Z}_2 = \{0, 1\}.$$

The nontrivial class (1) corresponds to a "large" gauge transformation.

1.3 SU(2) Doublet Fermions

Fermions charged under the weak force transform in the fundamental (doublet) representation **2** of $SU(2)$. A single such fermion field is a section of the associated vector bundle

$$\mathbf{2} \otimes S \longrightarrow M,$$

where S is the spinor bundle. For our purposes, the only important fact is:

A single SU(2) doublet fermion is one section of a vector bundle.

1.4 Witten's SU(2) Global Anomaly

A key result of Witten (1982) states that under a large gauge transformation $g : S^4 \to S^3$, the partition function of a single $SU(2)$ doublet fermion transforms as

$$Z_{\text{single doublet}} \longrightarrow (-1)^s \, Z_{\text{single doublet}},$$

where $s = [g] \in \pi_4(S^3) \simeq \mathbb{Z}_2$. Thus

$$s = 0 : \ Z \text{ invariant}, \qquad s = 1 : \ Z \mapsto -Z.$$

Therefore:

A single SU(2) doublet suffers from a global SU(2) anomaly.

1.5 Multiple Doublets and Anomaly Cancellation

If there are N doublets, the total transformation is

$$Z \longrightarrow (-1)^{Ns} Z.$$

The theory is gauge invariant for all gauge transformations if and only if N is even. In the Standard Model, one generation contains

$$3 \text{ quark doublets } + 1 \text{ lepton doublet } = 4 \text{ doublets}.$$

Hence

$$(-1)^{4s} = +1,$$

and the anomaly cancels.

This explains, for instance, why one cannot simply change $SU(3)$ color to $SU(4)$: it would change the number of weak doublets, potentially making N odd and producing an inconsistency.

1.6 Summary

- Large $SU(2)$ gauge transformations are classified by $\pi_4(S^3) \simeq \mathbb{Z}_2$.
- A single doublet fermion changes sign under the nontrivial transformation.
- A consistent theory must have an *even* number of doublets.
- The Standard Model satisfies this automatically ($N = 4$ per generation).

2 Fiber Choice in the Covering Map $\mathbb{R} \to S^1$

Consider the covering map
$$p : \mathbb{R} \longrightarrow S^1, \qquad p(x) = e^{2\pi i x}.$$
For any point $z = e^{2\pi i \alpha} \in S^1$, the fiber is

$$p^{-1}(z) = \{x \in \mathbb{R} \mid e^{2\pi i x} = z\} = \alpha + \mathbb{Z},$$

a translated copy of $\mathbb{Z}$. Thus every fiber is homeomorphic to the discrete set $\mathbb{Z}$.

In the long exact sequence of a fibration, one chooses a basepoint $b_0 \in S^1$ and defines the fiber as $p^{-1}(b_0)$. Although $b_0 = 1 \in S^1$ (the identity element) is the conventional choice, any b_0 works because S^1 is path-connected and homotopy groups based at different points are naturally isomorphic.

3 Homotopy Groups of the Discrete Space $\mathbb{Z}$

Although $\mathbb{Z}$ is not a manifold, it is a valid topological space and its homotopy groups are well-defined.

3.1 All Continuous Maps $S^n \to \mathbb{Z}$ Are Constant ($n \geq 1$)

Since $\mathbb{Z}$ is discrete, each singleton $\{k\} \subset \mathbb{Z}$ is open. A continuous map from a connected space S^n to a discrete space must have connected image, hence the image consists of only one point. Thus any map
$$f : S^n \to \mathbb{Z}, \quad n \geq 1$$
is constant.

3.2 Computation of $\pi_n(\mathbb{Z})$

If we fix the basepoint $k_0 \in \mathbb{Z}$, the only based map $S^n \to \mathbb{Z}$ is the constant map $f(x) \equiv k_0$. Hence, for $n \geq 1$,
$$\pi_n(\mathbb{Z}, k_0) = 0.$$

For $n = 0$, the path components of $\mathbb{Z}$ are just its points, so

$$\pi_0(\mathbb{Z}) \cong \mathbb{Z}.$$

3.3 Summary

$$\boxed{\pi_0(\mathbb{Z}) = \mathbb{Z}, \qquad \pi_{n\geq 1}(\mathbb{Z}) = 0.}$$

3.4 Application: Computing $\pi_n(S^1)$ Using the Covering Space

From the fibration

$$\mathbb{Z} \longrightarrow \mathbb{R} \longrightarrow S^1,$$

the long exact sequence gives

$$\pi_n(S^1) \cong \pi_{n-1}(\mathbb{Z}).$$

Using our calculation of $\pi_{n-1}(\mathbb{Z})$, we obtain the classical result:

$$\pi_n(S^1) = \begin{cases} \mathbb{Z}, & n = 1, \\ 0, & n \neq 1. \end{cases}$$

4 Understanding why $\partial : \mathbb{Z} \to \mathbb{Z}$ must be multiplication by 2

Consider the fibration

$$SO(2) \ \longrightarrow \ SO(3) \ \longrightarrow \ S^2.$$

The long exact sequence in homotopy (for $n = 3$ in the book's notation) contains the segment

$$\pi_2(SO(2)) \longrightarrow \pi_2(SO(3)) \longrightarrow \pi_2(S^2) \xrightarrow{\partial} \pi_1(SO(2)) \longrightarrow \pi_1(SO(3)) \longrightarrow \pi_1(S^2).$$

We substitute the known homotopy groups:

$$\pi_2(SO(2)) = 0, \quad \pi_2(S^2) = \mathbb{Z}, \quad \pi_1(SO(2)) = \mathbb{Z}, \quad \pi_1(SO(3)) = \mathbb{Z}_2, \quad \pi_1(S^2) = 0.$$

Thus the exact sequence becomes

$$0 \longrightarrow \pi_2(SO(3)) \longrightarrow \mathbb{Z} \xrightarrow{\partial} \mathbb{Z} \xrightarrow{\text{mod } 2} \mathbb{Z}_2 \longrightarrow 0.$$

4.1 Step 1: Determining the form of ∂

Exactness at $\pi_1(SO(2)) = \mathbb{Z}$ means

$$\mathrm{Im}(\partial) = \ker(\mathrm{mod}2).$$

Since

$$\ker(\text{mod}2) = 2\mathbb{Z},$$

we obtain

$$\text{Im}(\partial) = 2\mathbb{Z}.$$

Any group homomorphism $\partial : \mathbb{Z} \to \mathbb{Z}$ must have the form

$$\partial(n) = kn, \qquad k \in \mathbb{Z}.$$

Its image is $k\mathbb{Z}$. To satisfy $\text{Im}(\partial) = 2\mathbb{Z}$, the only possibility is

$$k = \pm 2.$$

Hence

$$\partial(n) = \pm 2n.$$

This is the meaning of the sentence in the book:

> "This forces $\partial : \mathbb{Z} \to \mathbb{Z}$ to be a multiplication by 2."

4.2 Step 2: Computing $\pi_2(SO(3))$

Exactness at $\pi_2(S^2) = \mathbb{Z}$ implies

$$\text{Im}\Big(\pi_2(SO(3)) \longrightarrow \mathbb{Z}\Big) = \ker(\partial).$$

Since $\partial(n) = \pm 2n$, its kernel is

$$\ker(\partial) = \{0\}.$$

Thus

$$\text{Im}\Big(\pi_2(SO(3)) \to \mathbb{Z}\Big) = 0.$$

But the map $0 \to \pi_2(SO(3))$ at the beginning of the sequence is injective, so the only way $\pi_2(SO(3))$ can map to 0 inside $\mathbb{Z}$ is if

$$\pi_2(SO(3)) = 0.$$

So the conclusion is:

$$\boxed{\pi_2(SO(3)) = 0.}$$